

Kinetic Energy derivation with initial angle and final angle

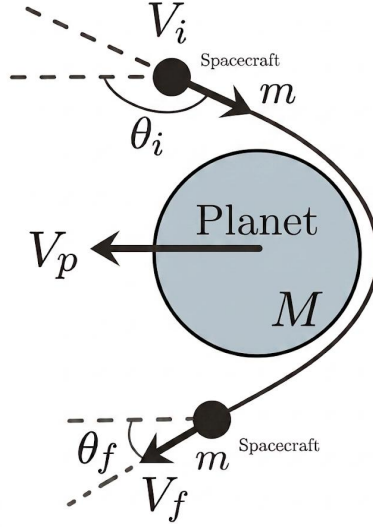


Figure 1: Diagram

Assumption: $M \gg m$, so the change in momentum and kinetic energy of the planet is negligible.

Let $\vec{v}_i'$ and $\vec{v}_f'$ be the initial and final velocity respectively in the planet frame. Then by conserving Kinetic Energy, we have:

$$\begin{aligned}
 |\vec{v}_i'|^2 &= |\vec{v}_f'|^2 \\
 |\vec{v}_i - \vec{v}_p|^2 &= |\vec{v}_f - \vec{v}_p|^2 \\
 (\vec{v}_i - \vec{v}_p) \cdot (\vec{v}_i - \vec{v}_p) &= (\vec{v}_f - \vec{v}_p) \cdot (\vec{v}_f - \vec{v}_p) \\
 |\vec{v}_i|^2 - 2|\vec{v}_i||\vec{v}_p|\cos\theta_i + |\vec{v}_p|^2 &= |\vec{v}_f|^2 - 2|\vec{v}_f||\vec{v}_p|\cos\theta_f + |\vec{v}_p|^2 \\
 |\vec{v}_f|^2 - 2|\vec{v}_f||\vec{v}_p|\cos\theta_f + 2|\vec{v}_i||\vec{v}_p|\cos\theta_i &= |\vec{v}_i|^2
 \end{aligned}$$

$$|\vec{v}_f| = |\vec{v}_p|\cos\theta_f + \sqrt{|\vec{v}_p|^2\cos^2\theta_f - 2|\vec{v}_i||\vec{v}_p|\cos\theta_i + |\vec{v}_i|^2}$$

The negative sign is ignored since magnitude cannot be negative.

For gravitational braking $|\vec{v}_i| > |\vec{v}_f|$

$$\Rightarrow |\vec{v}_i| - |\vec{v}_p| \cos \theta_f > \sqrt{|\vec{v}_p|^2 \cos^2 \theta_f - 2|\vec{v}_i||\vec{v}_p| \cos \theta_i + |\vec{v}_i|^2}$$

Square both sides and solve to get $\boxed{\cos \theta_f < \cos \theta_i}$ as the condition for gravitational braking

Derivation Analogous to Rutherford Scattering

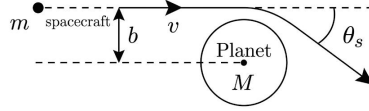


Figure 2: Diagram

→Solution 1 (Angular Momentum)

Assumption: $M \gg m$

Note that the entire question is solved in the planet frame

Let closest distance of approach = d

By conserving angular momentum:

$$mvb = mv_d d$$

$$\Rightarrow v_d = v \cdot \frac{b}{d}$$

By conserving energy:

$$\frac{1}{2}mv^2 = \frac{1}{2}mv_d^2 - \frac{GMm}{d}$$

So we have:

$$v_d^2 - v^2 = \frac{2GM}{d}$$

$$v^2 \cdot \left(\frac{b^2 - d^2}{d^2} \right) = \frac{2GM}{d}$$

$$d^2 + \left(\frac{2GM}{v^2} \right) d - b^2 = 0$$

$$d = -\frac{GM}{v^2} + \sqrt{\left(\frac{GM}{v^2}\right)^2 + b^2} = \frac{GM}{v^2} \left(\sqrt{1 + \left(\frac{bv^2}{GM}\right)^2} - 1 \right)$$

(Negative sign will be ignored because negative distance does not make physical sense)

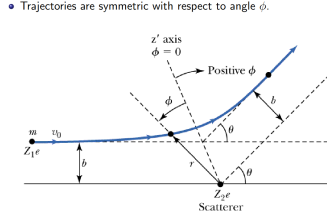


Figure 3: Diagram

Let $\vec{p}_i$ and $\vec{p}_f$ be the initial and final momentum of the spacecraft respectively.

We have $|\vec{p}_i| = |\vec{p}_f| = p$

Let $\Delta\vec{p} = \vec{p}_f - \vec{p}_i$ and scattering angle $= \theta_s$

ϕ is the angle the radial line to the spacecraft's position makes with the radial line to the closest position

$$\Delta\vec{p} = \int_{-\infty}^{\infty} \vec{F} dt$$

Since the trajectory is symmetric w.r.t $\phi = 0$

$$\Rightarrow |\Delta\vec{p}| = \int_{-\infty}^{\infty} |\vec{F}| \cos\phi dt = GMm \int_{-\infty}^{\infty} \cos\phi \frac{dt}{|\vec{r}|^2}$$

Where $\vec{F} = -\frac{GMm}{|\vec{r}|^3} \cdot \vec{r}$

Now angular momentum

$$\begin{aligned} \vec{L} &= \vec{r} \times \vec{p} = m\vec{r} \times \vec{v} = m\vec{r} \times \left(\frac{d\vec{r}}{dt} + |\vec{r}| \frac{d\vec{\phi}}{dt} \right) \\ \Rightarrow |\vec{L}| &= m|\vec{r}|^2 \cdot \frac{d\phi}{dt} \end{aligned}$$

But from initial conditions we have $|\vec{L}| = m v b$
 So by conserving angular momentum, we get:

$$\frac{dt}{|\vec{r}|^2} = \frac{d\phi}{v b}$$

So from the earlier equation get:

$$|\Delta \vec{p}| = \frac{G M m}{v b} \int_{-\phi_o}^{+\phi_o} \cos \phi \, d\phi$$

We know that $2\phi_o - \theta_s = \pi$

$$\Rightarrow \phi_o = \frac{1}{2}(\pi + \theta_s)$$

$$|\Delta \vec{p}| = \frac{2 G M m}{v b} \cdot \sin\left(\frac{1}{2}(\theta_s + \pi)\right)$$

From the triangle law for vectors, we have $|\Delta \vec{p}| = 2p \cdot \sin\left(\frac{\theta_s}{2}\right)$

$$2p \cdot \sin\left(\frac{\theta_s}{2}\right) = \frac{2 G M m}{v b} \cdot \cos\left(\frac{\theta_s}{2}\right)$$

$$\boxed{\theta_s = 2 \tan^{-1}\left(\frac{G M m}{2 K b}\right) = 2 \tan^{-1}\left(\frac{G M}{b v^2}\right)}$$

Where K = initial Kinetic energy

→Solution 2 (Hyperbolic Trajectory)

Since Energy > 0 , the trajectory is hyperbolic

Closest distance(d) = $a(e - 1)$ $\sin \phi = \frac{b}{a e}$

$$r = \frac{a(e^2 - 1)}{1 + e \cdot \cos \phi}$$

$$v_r = \dot{r} \hat{r} + r \dot{\theta} \hat{\theta}$$

$$a_r = (\ddot{r} - r \dot{\theta}^2) \hat{r}$$

$$\Rightarrow F(r) = m a_r = m(\ddot{a} - r \dot{\theta}^2) \hat{r}$$

Let $\frac{1}{u} = r$, then:

$$d\theta = b v u^2 dt$$

$$\dot{r} = -\frac{1}{u^2} \cdot \frac{du}{dt} = -bv \cdot \frac{du}{d\theta}$$

$$\ddot{r} = -(bv)^2 \cdot \frac{d^2u}{d\theta^2}$$

$$\Rightarrow F(1/u) = -GMmu^2 = -m(bv)^2 \left(\frac{d^2u}{d\theta^2} + u \right)$$

So we get Result 1:

$$\boxed{\frac{d^2u}{d\theta^2} + u = \frac{GM}{b^2v^2}}$$

From here we can say:

$$u = A_o \cdot \cos(\theta - \theta_o) + \frac{GM}{b^2v^2}$$

$$\Rightarrow r = \frac{\frac{b^2v^2}{GM}}{\frac{A_o b^2v^2}{GM} \cdot \cos(\theta - \theta_o) + 1} = \frac{a(e^2 - 1)}{1 + e \cdot \cos\theta}$$

$r_{min} = d$ occurs at $\cos(\theta - \theta_o) = 1$ and $\theta = 0$

$\Rightarrow \theta_o = 0$

$\Rightarrow e = \frac{A_o b^2v^2}{GM}$ and $d = \frac{\frac{b^2v^2}{GM}}{\frac{A_o b^2v^2}{GM} + 1}$

From here find d using the same method as the above solution:

$$\boxed{d = -\frac{GM}{v^2} + \sqrt{\left(\frac{GM}{v^2}\right)^2 + b^2} = \frac{GM}{v^2} (\sqrt{1 + \left(\frac{bv^2}{GM}\right)^2} - 1)}$$

Compare this to the value of d we got earlier to get:

$$A_o = \frac{1}{b} \sqrt{\left(\frac{GM}{bv^2}\right)^2 + 1}$$

Result 2:

$$\boxed{e = \sqrt{\left(\frac{GM}{bv^2}\right)^2 + 1}}$$

We knew that $\sin\phi = \frac{b}{ae}$, from that we get:

$$\phi = \tan^{-1}\left(\frac{bv^2}{GM}\right)$$

Deflection angle $(\delta) = \pi - 2\phi$, so Result 3:

$$\delta = 2\tan^{-1}\left(\frac{GM}{bv^2}\right)$$

→Solution 3 (Geometric Solution)

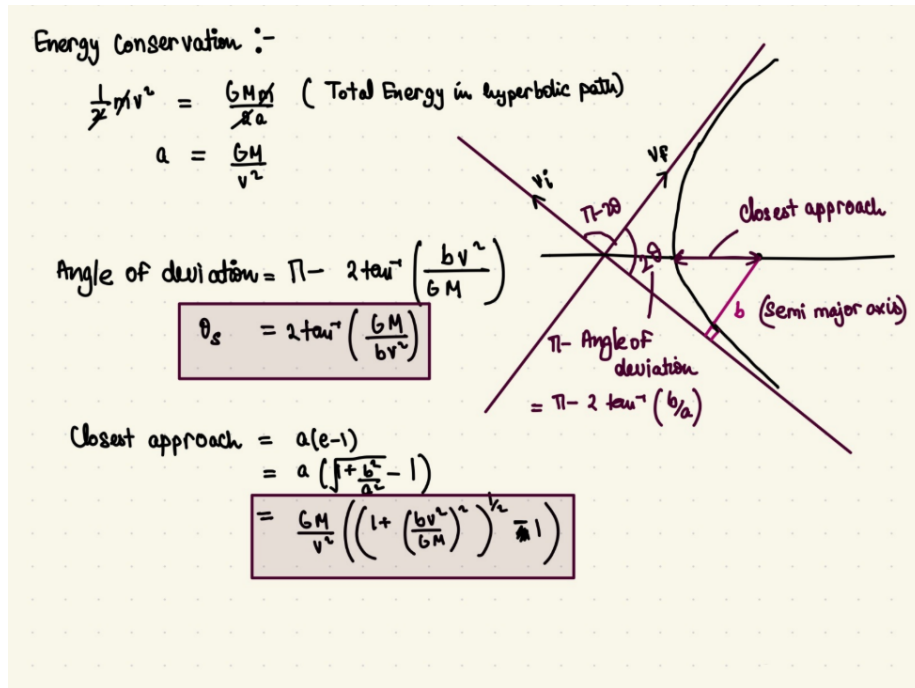


Figure 4: Diagram